

Supervised
 input data $x \rightarrow$ Function F $\xrightarrow{\sim}$ "desired output" T

Start with lot of examples S

Function has unknown parameters

Use examples to adjust parameters to minimize these "error"

$$F(x) - T$$

Neural net is a function:

x input

y desired output

$$x \rightarrow \sigma(A, x)$$

A , matrix x

σ simple function applied to each entry

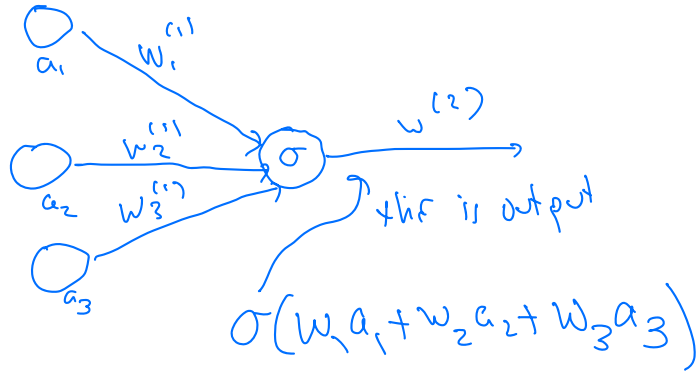
($\sigma = \frac{1}{1+e^{-x}}$, this is logistic regression)

$$\sigma_3(A_3 \sigma_2(A_2 \sigma_1(A_1 x)))$$

$$\sigma(x) = \begin{cases} x & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases} \quad \text{Relu}$$

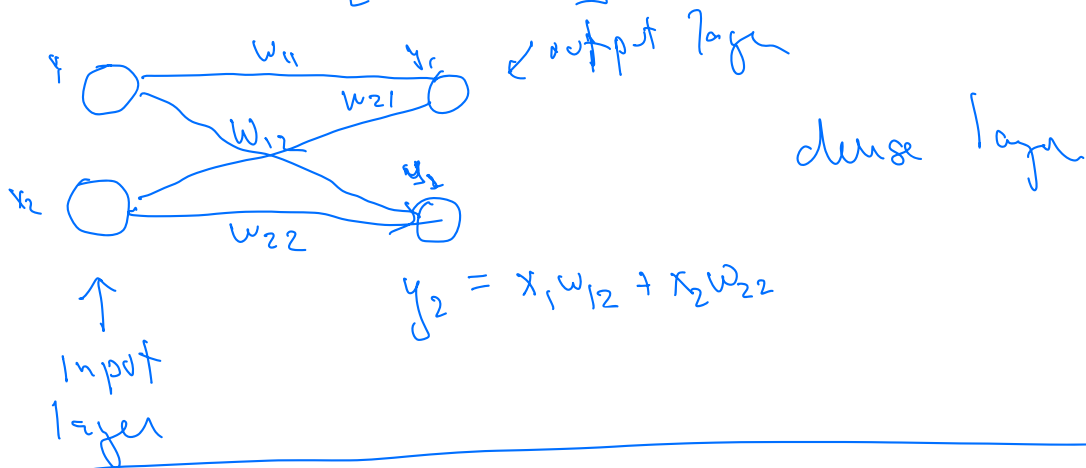
$$\sigma(x) = \frac{1}{1+e^{-x}}$$

$$\sigma(x) = \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$



input vector

$$\begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \end{bmatrix} = \begin{bmatrix} y_1 & y_2 \end{bmatrix}$$

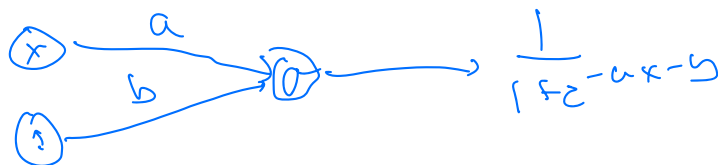


Simple Logistic Regression.

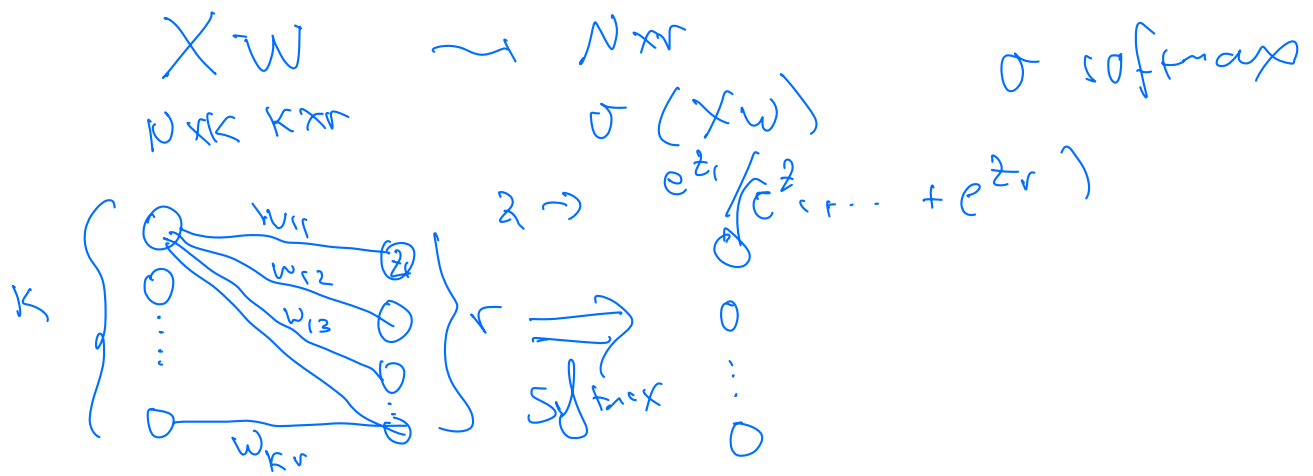
$$X \rightarrow X \begin{bmatrix} a \\ b \end{bmatrix} \xrightarrow{\sigma} \frac{1}{1 + e^{-ax-b}} \sim \frac{1}{1 + e^{-ax-b}}$$

↑
input
 $N \times 2$
second column is all ones

$$\begin{bmatrix} x & 1 \end{bmatrix}$$

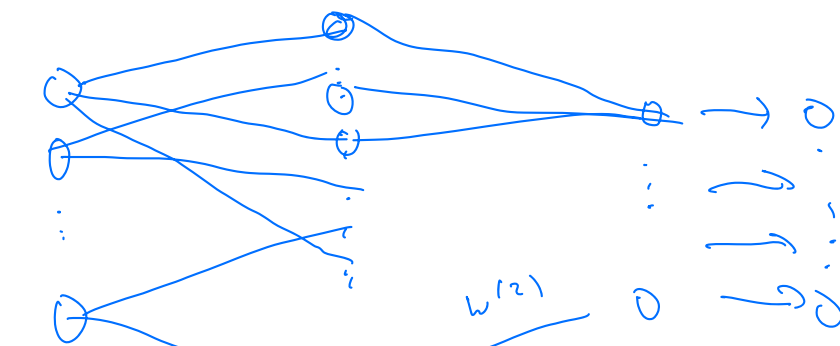


multiclass



perceptron

hidden layer



input layer
nodes
 $d^{(0)}$

$W^{(1)}$
 $d^{(0)} \times d^{(1)}$
matrix

$W^{(2)}$
 $d^{(1)} \times d^{(2)}$
matrix

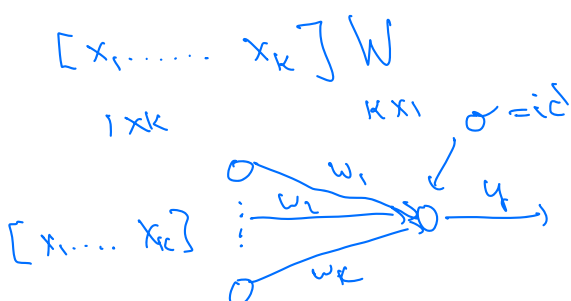
nodes $\rightarrow$ entry by entry

$$\sigma_2(\sigma_1(XW^{(1)})W^{(2)})$$

O.L.S. as a neural net

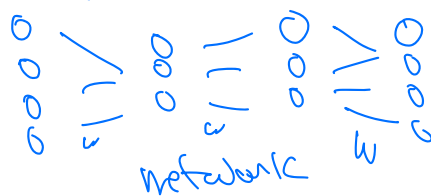
$$y \sim x_1w_1 + \dots + x_kw_k$$

$\sigma = id.$



$$\|y - y_{\text{desired}}\|^2 = \text{Loss}$$

Training data, target F (x^i, y^i) $\leftarrow$ learning target



"Loss" $\sim$ error between target

$$L \quad \text{compute } \sum_i L(F(x^i), y^i) = L$$

compute ∇L w.r.t weights in F
update weights using gradient

Repeat

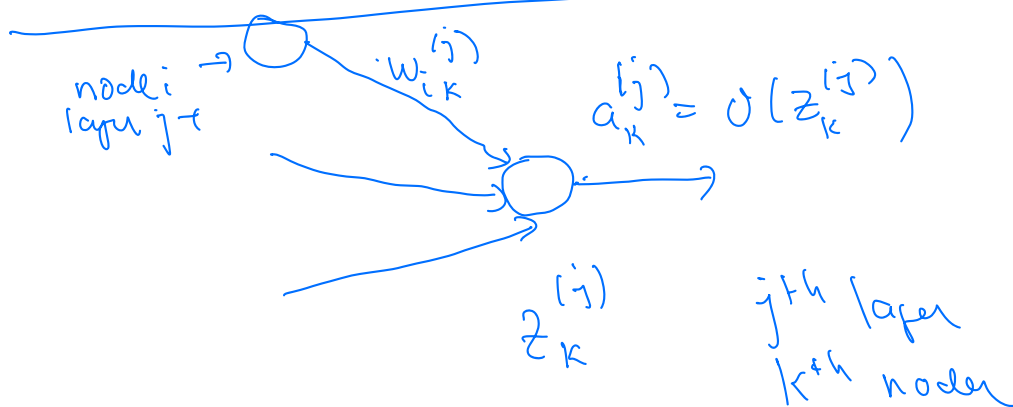
1 time through = training epoch

Weights $\leftrightarrow w_{ij}^{(k)}$

k layer
connect node i
in Layer $k-1$
to node j in layer k

∇

$$w_{ij}^{(k)} \rightarrow w_{ij}^{(k)} - \lambda \frac{\partial L}{\partial w_{ij}^{(k)}} \quad \text{learning rate}$$



$$z_k^{(j)} = \sum_i a_i^{(j-1)} w_{ik}$$

$$a_i^{(j-1)} = \sigma(z_i^{(j-1)})$$

Find

$$\delta_k^{(j)} = \frac{\partial L}{\partial z_k^{(j)}}$$

What is relationship between $\delta_k^{(j-1)}$ and $\delta_k^{(j)}$?

$$\delta_k^{(j-1)} = \frac{\partial L}{\partial z_k^{(j-1)}} = \sum_i \frac{\partial L}{\partial z_i^{(j)}} \frac{\partial z_i^{(j)}}{\partial z_k^{(j-1)}}$$

Side remark

$$F(x_1, x_2)$$

$$x_1 = x_1(t)$$

$$y_1 = y_1(t)$$

$$\frac{\partial F}{\partial t} = \frac{\partial F}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial F}{\partial y} \frac{\partial y}{\partial t}$$

$$z_i^{(j)} = \sum_t a_t^{(j-1)} w_{ti}^{(j)}$$

$$a_t^{(j-1)} = \sigma(z_t^{(j-1)})$$

$$z_{ki}^{(j)} = \sum_t \sigma(z_t^{(j-1)}) w_{ti}^{(j)}$$

$$\delta_k^{(j-1)} = \frac{\partial z_i^{(j)}}{\partial z_k^{(j-1)}} = \sigma'(z_k^{(j-1)}) w_{ki}^{(j)}$$

$$\delta_k^{(j-1)} = \sum_i \delta_i^{(j)} \sigma'(z_k^{(j-1)}) w_{ki}^{(j)}$$

$$\delta_k^{(j-1)} = \sigma'(z_k^{(j-1)}) w_{ki}^{(j)} \delta_i^{(j)}$$